

TURKU RAJAKARI, Finland

WMO: 027470

Lat: 60.378N

Lon: 22.096E

Elev: 6

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-14.2	-19.9	0.6	-17.2	-16.6	0.9	-13.5	16.1	1.1	14.5	1.0	4.3	60	0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	25.0	19.5	23.2	19.1	21.9	18.0	20.9	23.8	19.7	22.5	18.6	21.3	5.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	22.8	18.6	13.5	21.6	17.3	12.4	20.3	60.3	23.7	56.4	22.5	52.6	21.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.3	11.9	10.6		-18.7	27.6	5.0	2.5	-22.3	29.4	-25.2	30.9	-28.0	32.3	-31.6	34.1	
(5)			DB	-19.0	21.1	4.9	1.7	-22.6	22.3	-25.5	23.2	-28.2	24.2	-31.8	25.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-3.5	-3.2	-0.6	3.8	10.2	14.6	18.2	17.6	13.6	7.6	3.9	0.2
	DBStd	8.41	4.91	5.53	3.70	2.77	3.92	2.80	2.84	2.21	2.60	3.10	3.76	4.66
	HDD10.0	1926	418	368	328	188	46	1	0	0	5	86	182	303
	HDD18.3	4219	676	602	587	437	253	118	38	40	142	333	432	562
	CDD10.0	806	0	0	0	1	52	138	256	235	113	11	0	0
	CDD18.3	58	0	0	0	0	1	5	35	17	1	0	0	0
	CDH23.3	149	0	0	0	0	2	13	103	32	0	0	0	0
	CDH26.7	16	0	0	0	0	0	0	12	4	0	0	0	0
Wind	WSAvg	5.4	5.4	5.2	5.4	4.9	4.9	5.3	4.9	5.2	5.5	6.0	6.1	6.3
Precipitation	PrecAvg	659	52	34	39	37	31	48	56	86	58	76	71	62
	PrecMax	844	98	89	91	57	64	77	130	151	115	176	114	139
	PrecMin	487	3	0	8	5	4	9	0	31	25	10	9	15
	PrecStd	97	28	25	18	14	17	19	37	32	26	42	29	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.0	8.2	14.5	22.8	25.1	28.3	27.0	21.0	14.5	10.7	8.6
		MCWB	3.8	2.5	4.0	8.8	15.5	17.9	20.5	20.3	16.8	11.9	9.6	7.5
	2%	DB	4.0	3.9	6.0	11.6	20.0	22.5	25.9	23.7	18.9	13.2	9.9	7.2
		MCWB	3.1	2.6	3.3	6.8	14.0	17.3	19.8	19.1	15.9	11.8	9.3	6.1
	5%	DB	3.1	3.0	4.7	9.5	18.3	20.4	24.2	22.2	17.9	12.5	9.0	6.2
		MCWB	2.2	2.2	2.6	6.0	13.4	15.8	20.0	18.1	15.1	11.1	8.2	5.1
	10%	DB	2.2	2.2	3.7	7.6	16.4	18.9	22.7	20.8	17.1	11.7	8.1	5.4
		MCWB	1.5	1.6	2.1	4.8	12.6	14.8	19.2	17.2	14.8	10.3	7.2	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.2	5.0	9.3	16.4	19.3	22.1	21.8	17.5	12.8	9.9	7.7
		MCDB	4.7	4.0	6.7	13.6	20.3	23.3	25.3	24.5	19.6	13.4	10.2	8.3
	2%	WB	3.3	2.7	3.6	7.3	15.3	17.8	21.4	20.2	16.5	12.1	9.3	6.3
		MCDB	3.8	3.4	5.1	10.6	18.9	21.2	24.3	23.0	18.3	12.9	9.7	7.0
	5%	WB	2.4	2.2	2.9	6.2	14.2	16.7	20.5	19.0	15.7	11.5	8.4	5.2
		MCDB	3.0	2.9	4.4	8.6	17.5	19.8	23.4	21.4	17.4	12.3	9.0	6.1
	10%	WB	1.6	1.6	2.2	5.4	12.7	15.5	19.5	18.0	15.1	10.5	7.3	4.5
		MCDB	2.1	2.2	3.3	7.3	16.0	18.1	22.3	20.2	16.7	11.5	7.9	5.3
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.1	5.0	5.5	5.7	5.0	4.8	4.1	3.8	3.4	2.8	3.3
		MCDBR	2.9	3.6	5.0	8.6	9.0	7.8	6.9	5.8	4.8	3.1	3.0	3.6
		MCWBR	2.7	2.9	3.5	4.8	4.4	4.0	3.2	3.3	2.9	2.8	3.0	3.3
	5% WB	MCDBR	2.8	3.3	4.6	7.4	8.4	6.9	5.7	4.2	4.0	2.9	2.9	3.5
		MCWBR	2.9	2.9	3.5	4.4	4.5	4.1	3.1	3.1	2.8	2.9	3.0	3.3
Clear-Sky Solar Irradiance	taub	0.278	0.314	0.334	0.373	0.369	0.361	0.389	0.380	0.351	0.334	0.304	0.386	
	taud	2.156	2.237	2.310	2.317	2.404	2.457	2.401	2.421	2.488	2.397	2.211	2.608	
	Ebn at Noon	471	670	790	822	857	870	831	810	775	651	432	325	
	Edh at Noon	42	69	88	105	104	100	104	95	75	59	40	29	
All-Sky Solar Radiation	RadAvg	0.24	0.82	2.34	3.94	5.38	5.77	5.40	4.22	2.58	1.11	0.31	0.14	
	RadStd	0.04	0.16	0.28	0.38	0.50	0.43	0.51	0.41	0.31	0.17	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A

Nomenclature: See separate page